MSc Computing Dissertation · 2026

Predictive Maintenance and Equipment Failure Prediction Using the NASA Turbofan Engine Dataset

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Submitted: 2026

Word count: ~ 22,000

Abstract

Unscheduled equipment failure is one of the largest sources of operational cost in safety-critical industries such as aerospace, power generation, and advanced manufacturing. This dissertation investigates how multivariate sensor telemetry from the NASA C-MAPSS turbofan engine benchmark (FD001 subset) can be used to forecast Remaining Useful Life (RUL) and translate predictions into actionable maintenance decisions.

The methodology proceeds in six stages: data cleaning and RobustScaler preprocessing on 20,631 cycles across 21 sensors; feature engineering that expands 21 raw signals into 212 features (rolling mean/std, lag, delta, and cumulative); correlation-based feature selection; training of three regression models (Random Forest, XGBoost, and a multi-layer perceptron); evaluation on the held-out test set; and finally a four-tier maintenance decision matrix mapping predicted RUL to operator actions.

The Neural Network achieved the strongest results with MAE = 17.41, RMSE = 26.39, and R^2 = 0.5968, outperforming both tree-based ensembles. The resulting framework reduces unplanned downtime risk by surfacing imminent failures early while keeping healthy engines free of unnecessary preventive interventions.

1. Introduction

Modern industrial assets — turbofan engines, gas turbines, CNC machinery — are instrumented with hundreds of sensors that stream high-frequency telemetry. The cost of an unplanned shutdown typically dwarfs the cost of the sensor and analytics infrastructure required to anticipate it. Predictive maintenance converts raw telemetry into forward-looking intelligence: instead of servicing equipment on a calendar, operators service it when the data suggests failure is imminent.

This dissertation makes three contributions. First, it documents an end-to-end machine-learning pipeline applied to the public NASA C-MAPSS FD001 benchmark, producing reproducible artefacts (scaled data, engineered features, trained models, evaluation metrics). Second, it compares three contemporary regressors — Random Forest, XGBoost, and a multi-layer perceptron — under a common evaluation protocol. Third, it translates per-engine RUL forecasts into a four-tier maintenance decision matrix that practitioners can adopt without further statistical work.

2. Literature Review

The NASA C-MAPSS dataset was introduced by Saxena and Goebel (2008) and has become the de-facto benchmark for prognostic algorithms. Subsequent work — Saxena et al. (2008), Ramasso and Saxena (2014), Babu et al. (2016) — explored a wide range of approaches, from classical proportional-hazards models through ensemble trees to recurrent neural networks. Random Forests (Breiman, 2001) remain a strong baseline for tabular sensor data. XGBoost (Chen and Guestrin, 2016) introduces regularised gradient boosting that frequently wins on structured data. Deep models (LeCun, Bengio and Hinton, 2015) add representational capacity at the cost of larger training sets.

Recent comparative studies highlight that no single algorithm dominates predictive maintenance; performance is dominated by feature engineering choices and by the way time-dependence is encoded. The literature converges on the view that domain-aware feature construction — rolling statistics, lag windows, deltas, and cumulative features — is the largest single contributor to accuracy on the C-MAPSS benchmark.

3. Methodology

The pipeline implemented in this study comprises six phases:

1. Data cleaning and preprocessing (RobustScaler, missing-value audit, constant-column detection, outlier handling).
2. Feature engineering (rolling mean and std over windows of 3, 5, and 10 cycles; lag features at offsets 1, 2, 3, and 5; delta features; cumulative features).
3. Correlation-based feature selection ($|r| > 0.95$ deduplication).
4. Model training (Random Forest with 200 trees; XGBoost with 500 estimators and learning rate 0.05; MLP with hidden layers [256, 128, 64], ReLU, dropout, Adam).
5. Evaluation (MAE, RMSE, R^2 on the held-out 100-engine test set).
6. Maintenance decision matrix (four-tier mapping from RUL to action).

4. Implementation

Implementation was carried out in Python 3.14 with pandas, scikit-learn, XGBoost, and PyTorch. The training environment was a single workstation running the full pipeline in approximately 35 minutes. RobustScaler was used in place of StandardScaler because the data contains non-trivial outliers that represent genuine degradation phases rather than measurement error.

Feature engineering is performed inside an expanding-window groupby keyed by Unit_Number. The result is a 212-column feature matrix that is then split 80/20 for training and validation before being applied to the test set. All random seeds are fixed for reproducibility. Hyperparameters were selected by small grid search; final values are documented in Chapter 5.

5. Results & Discussion

Results on the 100-engine test set are summarised below.

Model	MAE	RMSE	R^2
Random Forest	21.84	31.27	0.4956
XGBoost	19.12	28.04	0.5572
Neural Net	17.41	26.39	0.5968

The Neural Network wins on every metric. Error is concentrated in mid-life engines (60–120 cycles of RUL) where individual sensor trajectories diverge sharply. The model is most accurate at the extremes (very healthy or very imminent failure), which is precisely where operational decisions are most valuable.

6. Maintenance Recommendation Framework

Predicted RUL is mapped to a four-tier decision matrix:

RUL range (cycles)	Action	Priority
> 120	Healthy — normal operation	Low
60 – 120	Schedule maintenance	Medium
30 – 60	Maintenance required	High
< 30	Immediate inspection	Critical

This mapping is deliberately conservative: the upper bound (120) matches the commonly-applied C-MAPSS RUL cap that prevents the model from wasting capacity on the uninformative early-life region. Each tier is paired with an operator-facing recommendation describing the next inspection step.

7. Conclusion & Future Work

This study demonstrates that a properly engineered pipeline — RobustScaler preprocessing, 212 time-series features, and a tuned multi-layer perceptron — predicts Remaining Useful Life on the NASA C-MAPSS FD001 benchmark with MAE 17.41 and R^2 0.5968, outperforming two strong tree-based baselines.

Future work will explore three directions. First, transfer learning from FD001 to the more challenging multi-condition subsets (FD002, FD003, FD004). Second, sequence models (LSTM, Temporal Convolutional Networks) that learn temporal structure directly rather than via hand-crafted features. Third, uncertainty quantification — predictive intervals rather than point estimates — to support risk-aware maintenance scheduling.

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